Experiment 9 – Automobile Speed Control using ANFIS Hybrid System

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1. Theory

- **ANFIS (Adaptive Neuro-Fuzzy Inference System)** merges fuzzy logic and neural networks.
- Fuzzy rules express human-like reasoning ("If slope ↑ and speed ↓ → increase throttle").
- Neural learning adjusts membership functions and rule weights automatically.
- The system learns input–output relations from data to maintain vehicle speed under varying road slopes and loads.
- Uses a **Sugeno-type FIS** trained by a **hybrid algorithm** (least-squares + back-propagation).

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2. Working (Detailed)

1. **Inputs (Slope Angle, Throttle %)**
   - Represent external driving conditions and driver's torque demand.
   - Provide real-time data that influence vehicle acceleration and control response.
2. **Error (e) and Change in Error (de)**
   - Error = (Desired Speed – Actual Speed); Change in Error shows how fast the error is varying.
   - These values form the control feedback that guides throttle adjustments.
3. **Scaling with Gains (Ke, Kde, Ku)**
   - Ke and Kde normalize the input signals to fit within fuzzy inference limits.
   - Ku scales the final output signal to suitable actuator range for throttle/brake control.

Symbol	Full Form	Acts On	Role	Formula
Ke	Error Gain	Error (e)	Normalizes speed difference	e(scaled) = Ke X e
Kde	Derivative Gain	Change of error (de/dt)	Normalizes rate of error change	de(scaled) = Kde X de/dt
Ku	Output Gain	Control signal (u)	Denormalizes fuzzy output	u(final) = Ku X u(fuzzy)

4. **ANFIS Controller Core (Fuzzy + Neural Integration)**

  - Fuzzy logic layer interprets linguistic rules such as “If error is small and decreasing → reduce throttle.”
  - Neural learning layer continuously tunes membership functions and rule weights for optimal behavior.

5. **ACO Optimization Loop (Metaheuristic Layer)**

  - Ant Colony Optimization fine-tunes Ke, Kde, Ku, λ, μ to minimize control error metrics (IAE, ISE, RMSE).
  - Ants explore multiple parameter paths and reinforce best-performing combinations through pheromone updating.

6. **Vehicle Dynamics Block (Plant Model)**

  - Receives control signal *u(t)* and converts it into vehicle speed response based on inertia and load.
  - Provides nonlinear simulation of real-world conditions like slope variation and friction.

7. **Feedback and Convergence**

  - The actual speed is continuously fed back to recalculate the new error.
  - Once minimal error and steady-state stability are achieved, final optimal parameters are fixed for the ANFIS controller.

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3. Block Diagram (with Explanation)

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Desired Speed ─┐
               │
               ▼
       [ Error (e) & ΔError (de) ]
               │
      (Scaled by Ke, Kde, Ku)
               │
     ┌────────────────────────────┐
     │      ANFIS Controller      │
     │  - Fuzzy Logic + NN Layer  │
     │  - ACO Optimization Loop   │
     └────────────────────────────┘
               │
      Engine Control Signal (u)
               ▼
     [ Vehicle Dynamics Model ]
   - Simulates speed response
   - Feeds back actual speed
               │
               ▼
      [ Feedback & Update ]
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    - Error recalculated
    - Parameters adjusted
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4. Advantages

- Automatically adapts to nonlinear and varying road conditions.
- Requires little manual tuning once trained.
- Provides smooth and stable speed response.
- Combines interpretability of fuzzy logic with learning power of neural nets.

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5. Disadvantages

- High computational cost during training.
- Performance depends on quality of training data.
- Choosing initial membership functions is critical.
- Real-time hardware implementation may be resource-intensive.
